

Cayley Product-Fiber Graphs of Finite Groups

Permutation Structure, Regular Actions, and Recursive Geometry

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September 2026

Abstract

Let $G = \{g_1, \dots, g_n\}$ be a finite group equipped with a linear ordering. For each $h \in G$ we consider the product fiber

$$F_h = \{(i, j) \in [n]^2 : g_i g_j = h\}$$

in the Cayley table. Each fiber is the graph of a permutation $\sigma_h \in S_n$, and the family of fiber permutations has a rigid algebraic form: if λ_h denotes the left regular permutation and τ the permutation induced by inversion, then

$$\sigma_h = \lambda_h \tau.$$

Consequently the family $\{\sigma_h : h \in G\}$ is a right coset of the left regular representation, and the regular representation can be reconstructed from the fibers by

$$\lambda_h = \sigma_h \sigma_e^{-1}.$$

We equip each fiber with an ordering-dependent local adjacency rule based on consecutive row or column indices. Under the natural identification of a fiber with $[n]$, its graph is the union of two Hamilton paths,

$$\Gamma_h = P_n \cup \sigma_h^{-1}(P_n),$$

which gives exact formulas for its number of edges and its cycle rank. We also describe the regular left action on fibers, a natural $G \times G$ left-right action, and continuous geometric realizations by vertex transport. The adjacency graph is not, in general, equivariant under these actions, so the geometric animation is formulated as continuous vertex transport together with a separate interpolation of the target edge layer. Finally, we develop two coupled recursive constructions generated by the fiber figures. The recursive cores form graph-directed iterated function systems; for a common contraction ratio r , their similarity dimension is determined by $nr^s = 1$ under the graph-directed open set condition. A permutation-labelled circle substitution encodes the group law locally through the rule $h \mapsto hg_j^{-1}$ at the j th vertex.

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1 Introduction

The multiplication table of a finite group is a Latin square: every group element occurs exactly once in each row and once in each column. This elementary fact makes it possible to replace the repeated symbols of the table by geometric objects. Instead of viewing an element h through a single cell or a single labelled vertex, we collect all positions at which the product equals h . The resulting set of positions is the *product fiber* of h .

A product fiber contains one point in each row and one point in each column, hence it is the graph of a permutation. This observation is the main algebraic entry point of the present construction. The resulting permutations are not arbitrary: they form a coset of the left regular representation. Thus the full regular permutation representation of G is already present in the relative position of the fibers.

To obtain sparse geometric figures, we connect two positions in the same fiber when their row indices or their column indices are consecutive. The resulting fibergraph is particularly rigid: after identifying the vertices by their column indices, it is the union of two Hamilton paths. This gives immediate combinatorial invariants and makes the dependence on the chosen ordering completely explicit.

The geometry also interacts naturally with multiplication. Left multiplication moves fibers regularly, and a left-right $G \times G$ action produces $h \mapsto xhy^{-1}$. The point sets are equivariant under these actions, while the ordering-dependent edge relation generally is not. We therefore distinguish carefully between the canonical algebraic motion of the vertices and the optional edge layer used for drawing.

A second theme is recursive geometry. Fiber figures can be inserted into one another, producing coupled substitution systems. The resulting recursive cores are graph-directed self-similar systems in the sense of Hutchinson and Mauldin–Williams [4, 5]. A permutation-labelled version uses the fiber permutation itself to decide which child figure occurs at each vertex, thereby encoding multiplication directly into the substitution rule.

The paper is organized so that the algebraic statements precede the visualization. This makes clear which features are intrinsic to the group, which are invariant only up to relabelling, and which depend essentially on the chosen linear order.

2 Ordered Cayley tables and product fibers

Let G be a finite group of order n , and fix an ordered list

$$G = (g_1, g_2, \dots, g_n).$$

Write $[n] = \{1, \dots, n\}$ and let

$$\nu : G \longrightarrow [n], \quad \nu(g_i) = i$$

be the index map associated with the order.

Definition 2.1 (Product fiber). For $h \in G$, define

$$F_h := \{(i, j) \in [n]^2 : g_i g_j = h\}.$$

The sets F_h partition the n^2 cells of the Cayley table.

Proposition 2.2 (Permutation form of a fiber). *For every $h \in G$ and every column $j \in [n]$, there is exactly one row index i with $(i, j) \in F_h$, namely*

$$i = \nu(hg_j^{-1}).$$

Consequently the map

$$\sigma_h : [n] \longrightarrow [n], \quad \sigma_h(j) := \nu(hg_j^{-1})$$

is a permutation and

$$F_h = \{(\sigma_h(j), j) : j \in [n]\}.$$

In particular, every fiber contains exactly n points, one in each row and one in each column.

Proof. For fixed h and j , the equation $g_i g_j = h$ is equivalent to $g_i = hg_j^{-1}$, which determines i uniquely. If two columns j and k had the same row value under σ_h , then $hg_j^{-1} = hg_k^{-1}$, hence $g_j = g_k$ and $j = k$. Thus σ_h is injective and therefore bijective. $\square$

The geometric realization used throughout this paper is the standard square-grid embedding

$$\phi(i, j) := (j, n + 1 - i) \in \mathbb{R}^2. \tag{1}$$

It places the first row at the top of the picture and the first column at the left.

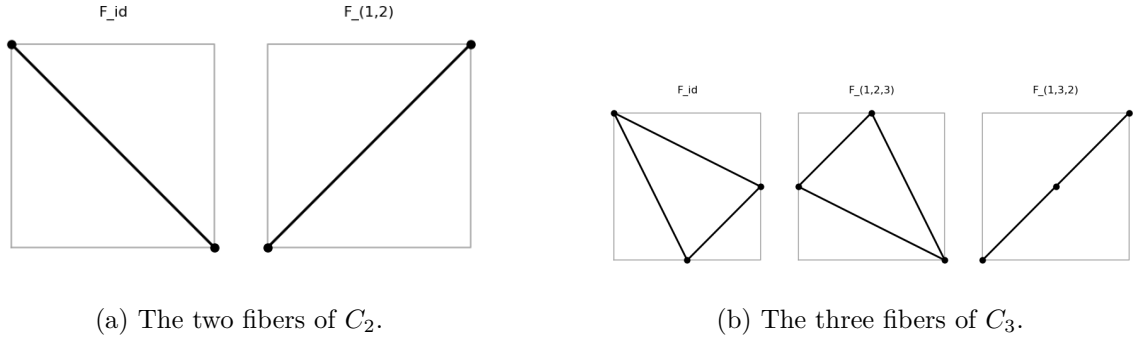


Figure 1: Product fibers become permutation patterns with a sparse local edge structure.

3 Fiber permutations and the regular representation

We now identify the exact algebraic structure of the permutations σ_h .

For $h \in G$, let

$$\lambda_h(i) := \nu(hg_i)$$

be the left regular permutation of h relative to the chosen order. Let

$$\tau(i) := \nu(g_i^{-1})$$

be the permutation induced by inversion.

Theorem 3.1 (Coset formula). *For every $h \in G$,*

$$\boxed{\sigma_h = \lambda_h \circ \tau.} \tag{2}$$

Hence

$$\boxed{\{\sigma_h : h \in G\} = \lambda(G)\tau,} \tag{3}$$

a right coset of the left regular subgroup $\lambda(G) \leq S_n$.

Proof. For $j \in [n]$,

$$(\lambda_h \circ \tau)(j) = \lambda_h(\nu(g_j^{-1})) = \nu(hg_j^{-1}) = \sigma_h(j).$$

The coset identity follows immediately. $\square$

Corollary 3.2 (Relative fibers recover the regular action). *For all $h, k \in G$,*

$$\boxed{\sigma_h \sigma_k^{-1} = \lambda_{hk^{-1}}.} \tag{4}$$

In particular,

$$\boxed{\lambda_h = \sigma_h \sigma_e^{-1}.} \tag{5}$$

Thus the family of product-fiber permutations determines the regular permutation representation of G .

Proof. Using theorem 3.1,

$$\sigma_h \sigma_k^{-1} = \lambda_h \tau (\lambda_k \tau)^{-1} = \lambda_h \tau \tau^{-1} \lambda_k^{-1} = \lambda_h \lambda_{k^{-1}} = \lambda_{hk^{-1}}.$$

Since $\sigma_e = \tau$, the special case $k = e$ gives (5). $\square$

Remark 3.3 (What is intrinsic). The individual coordinate vectors of the permutations depend on the chosen ordering, but the subgroup $\lambda(G)$ reconstructed from relative fiber permutations is the regular representation of G and changes only by conjugation when the ordering is changed; see section 5.

Example 3.4 (The group C_3). Let $C_3 = \{e, a, a^2\}$ with the order (e, a, a^2) . Then

$$\tau = [1, 3, 2], \quad \lambda_a = [2, 3, 1], \quad \lambda_{a^2} = [3, 1, 2],$$

and therefore

$$\sigma_e = [1, 3, 2], \quad \sigma_a = [2, 1, 3], \quad \sigma_{a^2} = [3, 2, 1].$$

The relation $\lambda_a = \sigma_a \sigma_e^{-1}$ is visible directly from these vectors.

4 The fibergraph is the union of two Hamilton paths

We now add the local adjacency used to turn each fiber into a sparse graph.

Definition 4.1 (Cayley product-fiber graph). Let

$$V = [n]^2.$$

Two distinct vertices $(i, j), (k, \ell) \in V$ are adjacent if and only if

$$g_i g_j = g_k g_\ell \quad \text{and} \quad (|i - k| = 1 \text{ or } |j - \ell| = 1). \quad (6)$$

The induced graph on F_h is denoted by Γ_h and called the *fibergraph* of h .

Let P_n denote the path on vertex set $[n]$ with edge set

$$E(P_n) = \{\{j, j+1\} : 1 \leq j < n\}.$$

Identify the vertex $(\sigma_h(j), j) \in F_h$ with its column label j .

Theorem 4.2 (Two-Hamilton-path theorem). *Under the column identification $F_h \cong [n]$,*

$$\boxed{\Gamma_h = P_n \cup \sigma_h^{-1}(P_n)}. \quad (7)$$

Equivalently, one Hamilton path visits the vertices in column order

$$1, 2, \dots, n,$$

and the second visits them in row order

$$\sigma_h^{-1}(1), \sigma_h^{-1}(2), \dots, \sigma_h^{-1}(n).$$

Proof. Two fiber vertices with column labels j and ℓ are adjacent by the column part of (6) exactly when $|j - \ell| = 1$, which gives P_n . They are adjacent by the row part exactly when

$$|\sigma_h(j) - \sigma_h(\ell)| = 1,$$

which is precisely the pullback $\sigma_h^{-1}(P_n)$. □

The theorem makes several graph invariants immediate. Define the adjacency statistic

$$a(\sigma) := \#\{j \in \{1, \dots, n-1\} : |\sigma(j+1) - \sigma(j)| = 1\}. \quad (8)$$

This counts the edges shared by the two Hamilton paths.

Corollary 4.3 (Edges, connectivity, and cycle rank). *For every $h \in G$,*

$$|V(\Gamma_h)| = n, \quad (9)$$

$$|E(\Gamma_h)| = 2(n-1) - a(\sigma_h), \quad (10)$$

$$\beta_1(\Gamma_h) = |E| - |V| + 1 = n - 1 - a(\sigma_h). \quad (11)$$

Moreover, Γ_h is connected and every vertex has degree at most 4.

Proof. Both paths have $n-1$ edges, and their intersection has exactly $a(\sigma_h)$ edges. Since P_n is a spanning subgraph, Γ_h is connected. The cycle-rank formula follows from $\beta_1 = |E| - |V| + 1$ for a connected finite graph. Each vertex has degree at most two in each Hamilton path. □

Corollary 4.4 (When is a fibergraph a tree?). *The graph Γ_h is a tree if and only if*

$$a(\sigma_h) = n - 1.$$

Equivalently, σ_h traverses the path P_n monotonically, so

$$\sigma_h = [1, 2, \dots, n] \quad \text{or} \quad \sigma_h = [n, n-1, \dots, 1].$$

Proof. By (11), the graph is a tree exactly when its cycle rank is zero. A permutation in which every pair of consecutive positions maps to adjacent values is a Hamilton traversal of P_n , and a path has only the two orientations shown. $\square$

Example 4.5 (Cycle ranks in C_3). For the permutations in theorem 3.4,

$$a(\sigma_e) = 1, \quad a(\sigma_a) = 1, \quad a(\sigma_{a^2}) = 2.$$

Hence the first two fibergraphs have three edges and cycle rank 1, while the third has two edges and is a path.

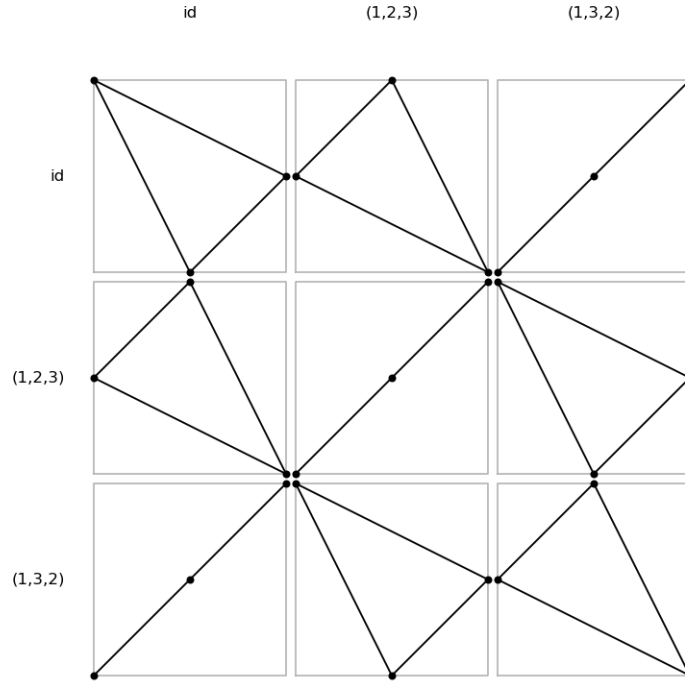


Figure 2: The Cayley table of C_3 rendered by product-fiber figures. Each cell contains the complete figure associated with its product value.

5 Dependence on the linear ordering

The point-fiber construction is algebraic, but the local adjacency rule (6) uses the numerical notion of consecutive indices. Therefore a linear ordering is genuine input data and must be treated explicitly.

5.1 How fiber permutations change under reordering

Let $\pi \in S_n$, and replace the index map ν by

$$\nu' = \pi \circ \nu.$$

Equivalently, the same group elements are relabelled simultaneously in rows and columns by π .

Proposition 5.1 (Simultaneous conjugation). *Under this reordering,*

$$\sigma'_h = \pi \sigma_h \pi^{-1}, \quad \lambda'_h = \pi \lambda_h \pi^{-1}, \quad \tau' = \pi \tau \pi^{-1}.$$

Hence the coset identity $\sigma_h = \lambda_h \tau$ is invariant under relabelling up to simultaneous conjugation.

Proof. If $j' = \pi(j)$, then

$$\sigma'_h(j') = \nu'(hg_j^{-1}) = \pi \nu(hg_j^{-1}) = \pi \sigma_h(j),$$

which gives the first identity. The other two are identical calculations. $\square$

The graph Γ_h , however, need not be preserved, because the distinguished path P_n is tied to the current numerical order. An arbitrary permutation π does not preserve adjacency in P_n .

5.2 Why lexicographic sorting of the regular action does not canonize the order

A tempting construction is to represent each $g \in G$ by the image vector

$$(\lambda_g(1), \dots, \lambda_g(n))$$

and lexicographically sort these vectors. The following proposition shows that this procedure does not remove the dependence on the initial ordering.

Proposition 5.2 (Degeneracy of regular lexicographic sorting). *Start from an ordered list $(g_1, \dots, g_n)$ and lexicographically sort the regular-action vectors*

$$(\nu(gg_1), \nu(gg_2), \dots, \nu(gg_n)), \quad g \in G.$$

The resulting order is exactly

$$\boxed{(g_1 g_1^{-1}, g_2 g_1^{-1}, \dots, g_n g_1^{-1})}. \quad (12)$$

In particular, if $g_1 = e$, the procedure returns the original ordering unchanged.

Proof. The first coordinate of the vector for g is $\nu(gg_1)$. As g varies over G , this coordinate assumes each value $1, \dots, n$ exactly once, so lexicographic comparison is decided by the first coordinate alone. The unique g with first coordinate k satisfies $gg_1 = g_k$, hence $g = g_k g_1^{-1}$. $\square$

There is also a conceptual obstruction to an automorphism-invariant total order.

Proposition 5.3 (No intrinsic automorphism-invariant order in general). *If a finite group G has a total order preserved by every automorphism of G , then $\text{Aut}(G)$ is trivial. Consequently, any group with a nontrivial automorphism group admits no total order that is simultaneously intrinsic and invariant under all automorphisms.*

Proof. Every order-preserving bijection of a finite total order fixes the least element, then the next element, and so on; hence it is the identity. Therefore every automorphism would have to be the identity. $\square$

For computations one should therefore state the ordering convention rather than call it canonical. Reasonable choices include: an explicitly chosen order; the lexicographic order of image vectors when $G \leq S_m$ is given as a concrete permutation group; or a canonical labelling produced by an external graph-isomorphism algorithm applied to a chosen encoding of the multiplication table [6]. The last option produces a reproducible computational representative, but it remains a computational convention rather than an automorphism-fixed order on the underlying abstract set.

6 Group actions on product fibers

6.1 The regular left action

For $x \in G$, define

$$T_x : F_h \longrightarrow F_{xh}, \quad T_x(i, j) := (\lambda_x(i), j).$$

Proposition 6.1 (Regular action on fibers). *The map T_x is a bijection $F_h \rightarrow F_{xh}$, and the induced action of G on the family $\{F_h : h \in G\}$ is isomorphic to the regular left action of G on itself. Moreover,*

$$\boxed{\sigma_{xh} = \lambda_x \sigma_h.} \tag{13}$$

Proof. If $(i, j) \in F_h$, then

$$g_{\lambda_x(i)} g_j = (x g_i) g_j = x h,$$

so $T_x(i, j) \in F_{xh}$. Bijectivity follows from the fact that λ_x is a permutation. Equation (13) follows either directly or from $\sigma_{xh} = \lambda_{xh} \tau = \lambda_x \lambda_h \tau$. $\square$

The action is canonical on the point fibers, but it need not act by graph isomorphisms on Γ_h . Column-consecutive edges are preserved because the column index does not move. Row-consecutive edges can be destroyed because a regular permutation λ_x need not preserve the path adjacency $|i - k| = 1$.

6.2 A natural left-right action

There is a larger symmetry on the product fibers. Define the right permutation

$$\rho_y(j) := \nu(g_j y^{-1}), \quad y \in G.$$

Then $y \mapsto \rho_y$ is a permutation representation and

$$T_{x,y}(i, j) := (\lambda_x(i), \rho_y(j)).$$

Theorem 6.2 ($G \times G$ action). *The maps $T_{x,y}$ define an action of $G \times G$ on the set of all table positions, and*

$$T_{x,y}(F_h) = F_{xhy^{-1}}. \tag{14}$$

The left regular action is the subgroup $G \times \{e\}$, the right regular action is $\{e\} \times G$, and the diagonal subgroup acts on fiber labels by conjugation:

$$(x, x) : h \longmapsto x h x^{-1}.$$

Proof. One has $\lambda_{x_1} \lambda_{x_2} = \lambda_{x_1 x_2}$ and $\rho_{y_1} \rho_{y_2} = \rho_{y_1 y_2}$, so the coordinate maps form a $G \times G$ action. If $(i, j) \in F_h$, then

$$g_{\lambda_x(i)} g_{\rho_y(j)} = (x g_i) (g_j y^{-1}) = x h y^{-1}.$$

The subgroup statements follow immediately. $\square$

Remark 6.3. The left-right action suggests two-dimensional animations in which row and column coordinates move simultaneously. The diagonal subgroup gives a geometric visualization of conjugacy dynamics, while the left subgroup gives the regular action used most naturally for a row-by-row multiplication display.

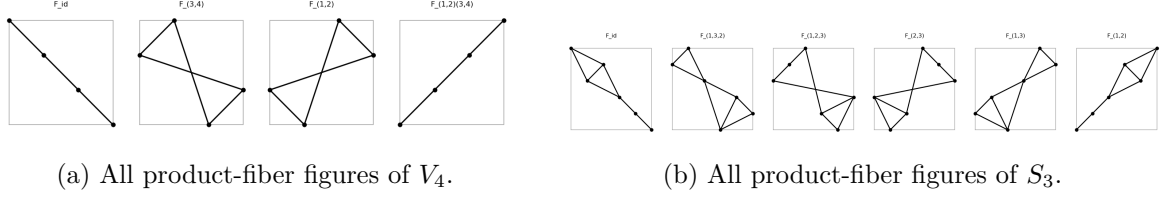


Figure 3: Fiber families for two groups of different orders and algebraic types.

7 Continuous geometric realization

The embedding (1) turns a fiber into a planar point configuration. For $(i, j) \in F_h$ write

$$p_{ij} = \phi(i, j) = (j, n + 1 - i).$$

Under the left action of x , the target point is

$$q_{ij}^{(x)} = \phi(\lambda_x(i), j) = (j, n + 1 - \lambda_x(i)).$$

Define the linear interpolation

$$p_{ij}(t) := (1 - t)p_{ij} + tq_{ij}^{(x)}, \quad 0 \leq t \leq 1. \quad (15)$$

Every vertex therefore moves vertically and continuously.

If a source edge connects p_{ij} and $p_{k\ell}$, its transported geometric segment at time t is

$$E_t = [p_{ij}(t), p_{k\ell}(t)].$$

This produces a continuous transport of the *source edge layer*. However, because the edge rule depends on consecutive indices, the transported source edges at $t = 1$ need not equal the actual edges of Γ_{xh} .

For a mathematically faithful animation it is therefore useful to keep two layers:

- (i) the transported source layer, whose endpoints move according to (15);
- (ii) the true target layer, consisting of the edges of Γ_{xh} at their fixed target positions.

Choose continuous opacity functions $\alpha, \beta : [0, 1] \rightarrow [0, 1]$ with

$$\alpha(0) = 1, \quad \alpha(1) = 0, \quad \beta(0) = 0, \quad \beta(1) = 1.$$

The rendered scene is the moving source layer with opacity $\alpha(t)$ together with the target layer with opacity $\beta(t)$. This is a continuous geometric visualization of the canonical vertex action without falsely asserting that the ordering-dependent graph structure itself is equivariant.

For the $G \times G$ action of theorem 6.2, the target of (i, j) is

$$\phi(\lambda_x(i), \rho_y(j)),$$

so the same interpolation gives a two-dimensional motion of vertices.

7.1 Row-by-row action display

Fix the ordered group elements $g_1, \dots, g_n$ and an actor $x \in G$. Display the source figures

$$F_{g_1}, F_{g_2}, \dots, F_{g_n}$$

in a top row and their images

$$F_{xg_1}, F_{xg_2}, \dots, F_{xg_n}$$

in a bottom row. Each column then records one value of the regular action. Running through all actors $x \in G$ gives an animated dictionary of the multiplication law.

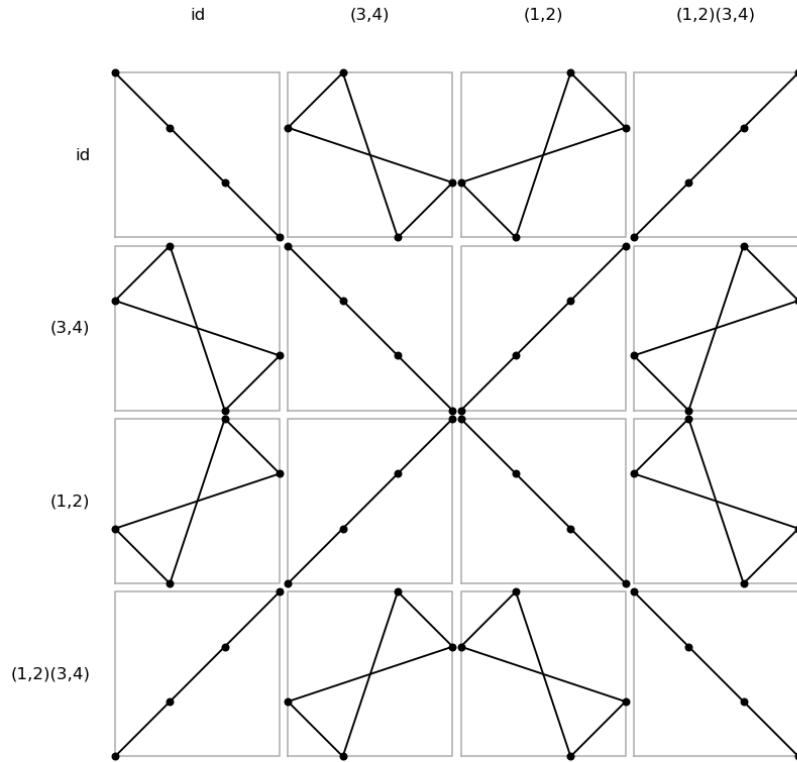


Figure 4: The Cayley table of V_4 rendered with the fiber figure associated with each product.

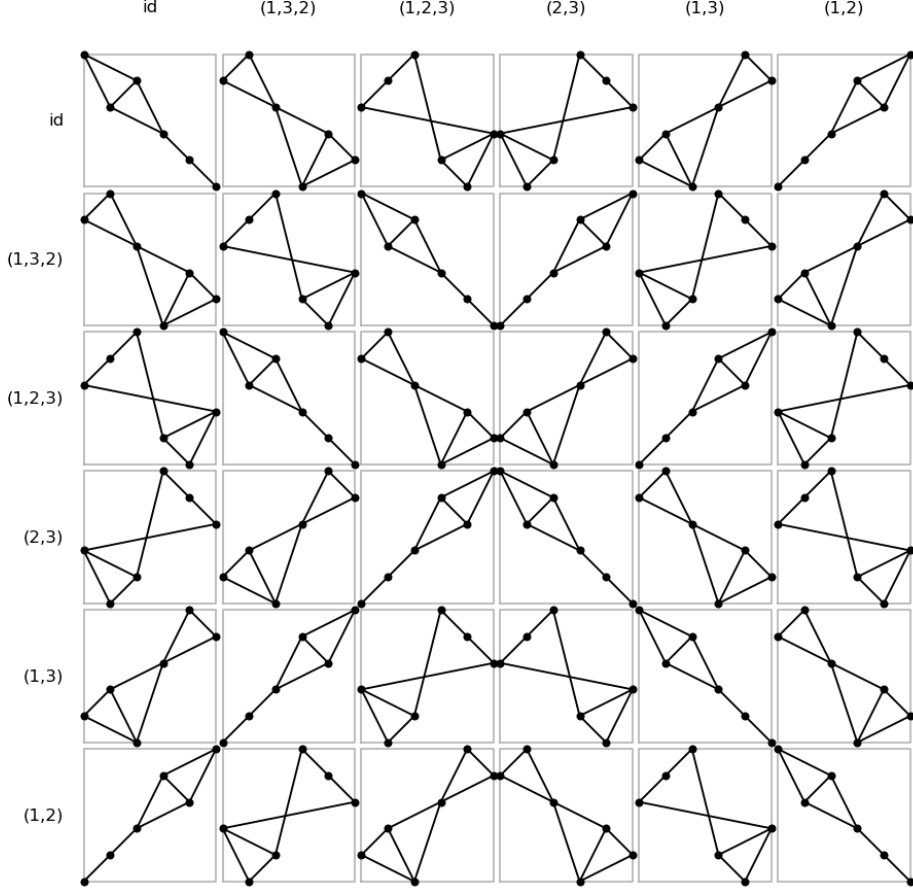


Figure 5: The same rendering for S_3 . The visual complexity reflects the ordering-dependent permutations σ_h , while the relative permutations still recover the regular representation.

8 Coupled recursive geometry

We next use the fiber figures as generators of recursive systems. It is useful to separate the *recursive core*, determined only by contractive copies, from additional connector edges or circles used for visualization.

8.1 Uniform coupled substitution

Let $\mathcal{F} = \{F_g : g \in G\}$ be a family of normalized geometric models, one for each group element. For every outer state $h \in G$ and every vertex index $j \in [n]$, choose a contracting similarity

$$\phi_{h,j} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

with common contraction ratio $0 < r < 1$, placing a small copy near the j th vertex of the outer figure. Define compact sets recursively by the graph-directed equations

$$K_h = \bigcup_{j=1}^n \phi_{h,j}(K_{g_j}). \quad (16)$$

Thus every state contains one copy of every state.

The directed substitution graph has vertex set G and exactly one directed edge from every state to every state. Its adjacency matrix is the all-ones matrix J_n , whose spectral radius is n .

Theorem 8.1 (Similarity dimension of the uniform recursive core). *Assume the similarities in (16) all have ratio r and satisfy the graph-directed open set condition. Then every strongly connected component is the whole state graph, and the common Hausdorff dimension s of the recursive cores is the unique solution of*

$$nr^s = 1.$$

Hence

$$s = \frac{\log n}{-\log r}. \quad (17)$$

Proof. For a graph-directed similarity system, the weighted adjacency matrix at exponent s is $r^s J_n$. Its spectral radius is nr^s . Under the graph-directed open set condition, the Hausdorff dimension is the exponent for which this spectral radius equals 1 [5]. $\square$

Remark 8.2. The formula depends only on the group order n and the common ratio r . Thus groups of the same order have the same similarity dimension in this uniform scheme. Their differences occur in the geometry of the placements, overlaps, and connector structures rather than in the all-ones substitution matrix itself.

A computational variant connects child copies corresponding to adjacent outer vertices by choosing the nearest pair of child vertices, with a deterministic tie-breaking rule. This yields visually coherent finite approximants, but the connector selection depends on the geometry at the current scale. For a mathematically local recursive rule, the permutation-labelled circle construction below is preferable.

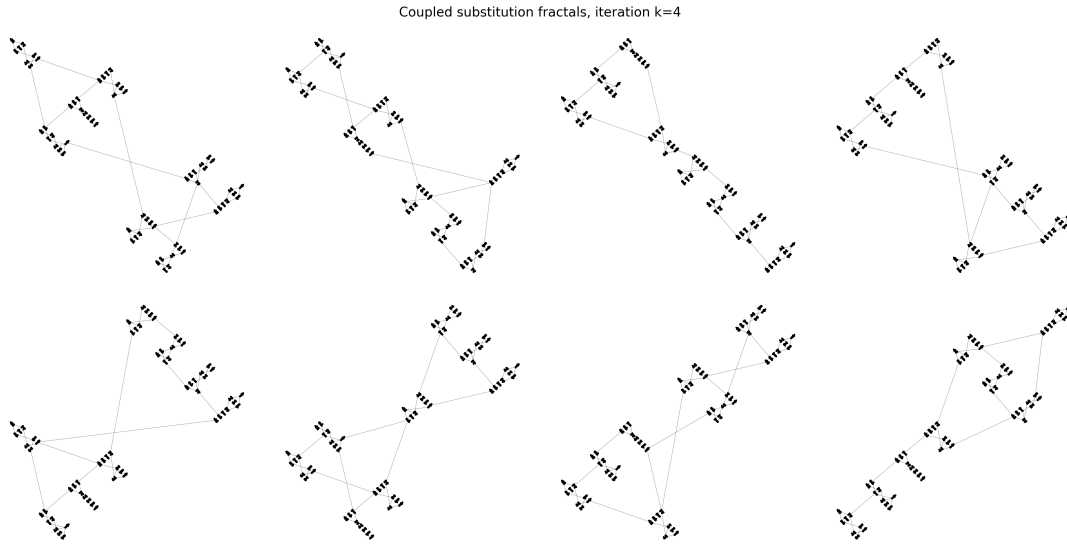


Figure 6: A gallery of coupled substitution approximants for the quaternion group Q_8 . The recursive placement is coupled across the family rather than being a simple self-copy of a single figure.

Coupled substitution fractal of F_{id} ($k=4$)

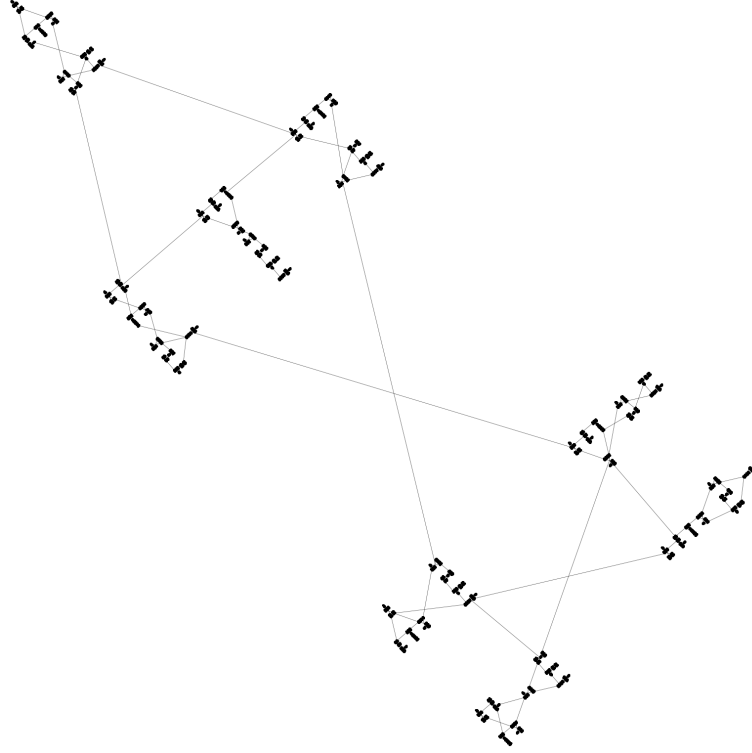


Figure 7: A fourth-stage coupled substitution approximant associated with the identity fiber of Q_8 .

8.2 Permutation-labelled circle substitution

The fiber permutation gives a natural local rule that inserts child figures according to the group law itself.

For each $h \in G$, enumerate the vertices of F_h from left to right, i.e. by their column index j . In table coordinates the j th vertex is

$$p_h(j) = (\sigma_h(j), j).$$

By definition of σ_h ,

$$\boxed{g_{\sigma_h(j)} = hg_j^{-1}}. \quad (18)$$

Thus the row label of the j th fiber point is exactly the group element hg_j^{-1} .

Fix a radius parameter $\rho > 0$. Around each vertex $p_h(j)$ place a disk

$$C_h(j) = \{x \in \mathbb{R}^2 : \|x - p_h(j)\| \leq \rho\}.$$

Inside this disk insert a normalized copy of the child figure

$$\boxed{F_{hg_j^{-1}}}. \quad (19)$$

Equivalently, the child index is $\sigma_h(j)$.

To normalize a geometric figure F_m with vertex set $P_m \subset \mathbb{R}^2$, choose its barycenter

$$c_m = \frac{1}{n} \sum_{q \in P_m} q$$

and radius

$$R_m = \max_{q \in P_m} \|q - c_m\|.$$

The map

$$q \mapsto \frac{q - c_m}{R_m}$$

centers the figure at the origin and places its vertices in the unit disk. If $0 < \lambda \leq 1$ is a filling parameter, the copy placed into the disk centered at p has the form

$$q \mapsto p + \lambda \rho \frac{q - c_m}{R_m}. \quad (20)$$

For the outer connector layer, retain the edges of the outer fibergraph rather than recomputing nearest neighbours between recursively expanded copies. If the outer graph has an edge between vertices $p_h(a)$ and $p_h(b)$, put

$$u_{ab} = \frac{p_h(b) - p_h(a)}{\|p_h(b) - p_h(a)\|}$$

and draw the clipped segment

$$[p_h(a) + \rho u_{ab}, p_h(b) - \rho u_{ab}]. \quad (21)$$

The segment points toward the centers of the two disks and stops at their boundaries.

Definition 8.3 (Circle-substitution approximants). Set $C_h^{(0)} = F_h$. Inductively, $C_h^{(k+1)}$ is obtained from the outer template F_h by replacing its j th vertex with a disk containing a scaled copy of

$$C_{hg_j^{-1}}^{(k)}$$

using (20), and by drawing the clipped outer connectors (21).

Proposition 8.4 (Locality and completeness of the child rule). *For every outer state h , the list of children*

$$(hg_1^{-1}, hg_2^{-1}, \dots, hg_n^{-1})$$

contains every element of G exactly once. The child attached to a vertex is determined locally by the outer fiber permutation and does not require any global nearest-neighbour search.

Proof. Right multiplication by inversion, $g_j \mapsto hg_j^{-1}$, is a bijection of G . Identity (18) shows that this list is exactly the row-label sequence encoded by σ_h . $\square$

Thus the directed state graph is again complete, but the *spatial placement* of the states is group-dependent.

8.3 Dimension of the recursive core and of the decorated construction

Assume the normalized recursive copies in (20) are similarities with common ratio

$$r = \lambda \rho.$$

Under the graph-directed open set condition, the recursive core again has dimension

$$s = \frac{\log n}{-\log(\lambda \rho)}. \quad (22)$$

A convenient scale is

$$\rho = \frac{1}{2n}.$$

Then

$$s = \frac{\log n}{\log(2n/\lambda)}. \quad (23)$$

For $0 < \lambda \leq 1$ and $n \geq 2$, one has $s < 1$.

It is important to distinguish this recursive core from the decorated object that also contains circles and connector segments. Let D_h be the closure of the union of the recursive core together with all circle boundaries and clipped connector segments appearing at all finite levels.

Proposition 8.5 (Dimension of the decorated limit). *Assume $n \geq 2$, $0 < \lambda \leq 1$, $\rho = 1/(2n)$, and the graph-directed open set condition for the recursive core. If the connector layer contains at least one nondegenerate segment, then*

$$\dim_H D_h = 1.$$

Proof. The recursive core has dimension $s < 1$ by (23). At each finite level there are only finitely many circle boundaries and line segments. Over all levels these form a countable union of sets of Hausdorff dimension 1, so their union has Hausdorff dimension at most 1. Since a nondegenerate segment is present, the dimension is at least 1. Adding the recursive core cannot increase the maximum beyond 1. $\square$

This distinction prevents a common ambiguity: the exponent in (23) describes the recursively nested copy set, not the dimension of a drawing in which one-dimensional connectors are retained at all scales.

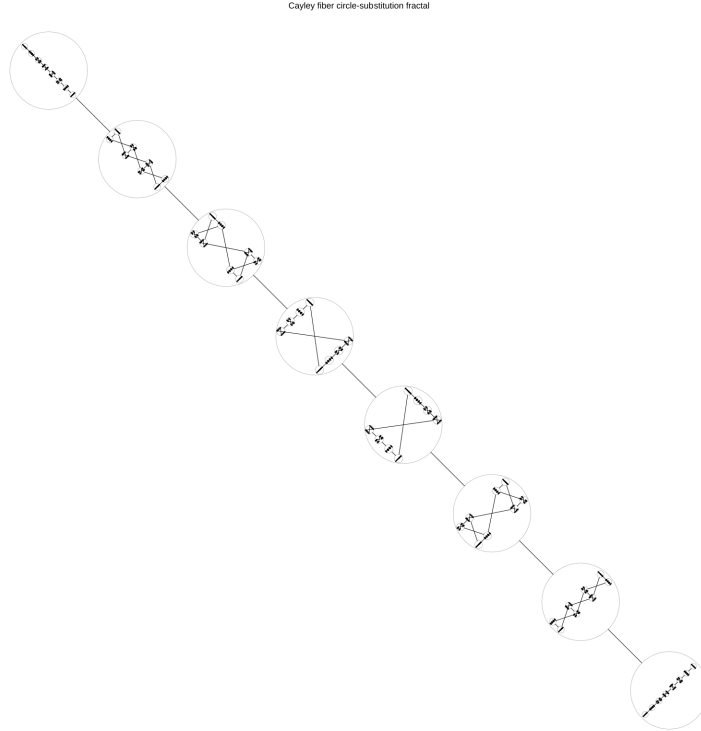


Figure 8: A circle-substitution approximant for the elementary abelian group C_2^3 . The circles localize the recursive copies, while clipped outer connectors preserve the combinatorics of the parent fibergraph.

9 Larger examples

The construction scales directly with the group order. The following examples illustrate how the same algebraic rules produce increasingly complex fiber families.

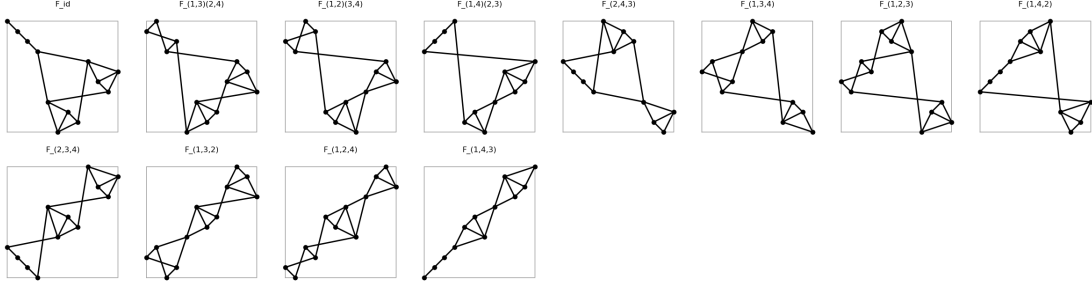


Figure 9: All twelve product-fiber figures of A_4 in one fixed ordering. The figures are ordering-dependent, but their relative permutations recover the left regular representation.

10 Computational construction

For a finite group represented in a computer algebra system, the basic data can be produced without any geometric heuristics.

Fiber permutation algorithm

Given an ordered list $G = (g_1, \dots, g_n)$ and the index dictionary ν , for each $h \in G$ compute

$$\sigma_h(j) = \nu(hg_j^{-1}), \quad j = 1, \dots, n.$$

The fiber points are $(\sigma_h(j), j)$. The edge set can then be obtained in linear time from the two path descriptions:

$$\begin{aligned} E_{\text{col}} &= \{\{j, j+1\} : 1 \leq j < n\}, \\ E_{\text{row}} &= \{\{\sigma_h^{-1}(r), \sigma_h^{-1}(r+1)\} : 1 \leq r < n\}. \end{aligned}$$

Thus one does not need to test all $\binom{n}{2}$ pairs of fiber vertices.

Action algorithm

For an actor x , compute the regular permutation

$$\lambda_x(i) = \nu(xg_i).$$

Then the target permutation of each fiber is obtained by

$$\sigma_{xh} = \lambda_x \sigma_h,$$

which provides both an algebraic consistency check and the target point configuration for animation.

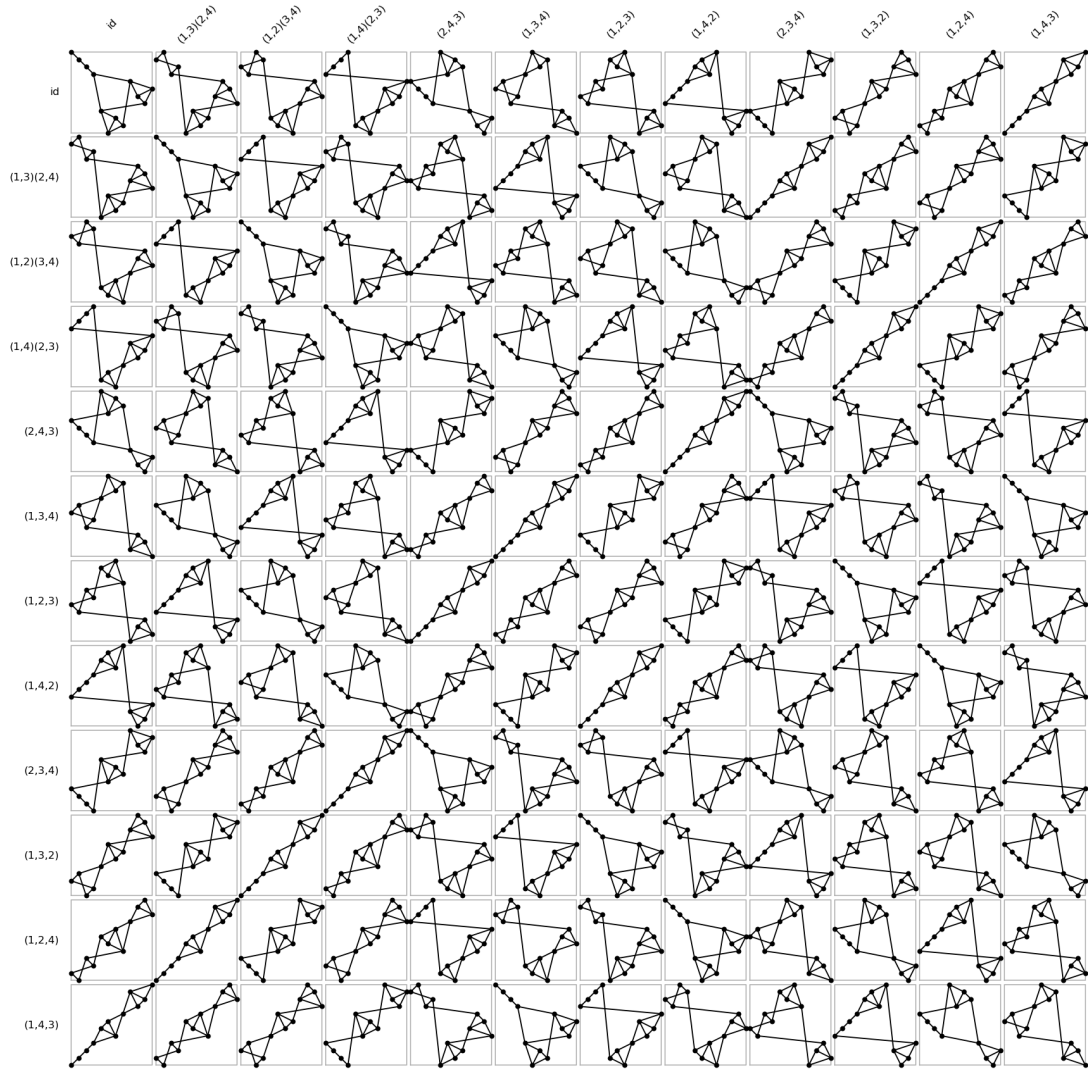


Figure 10: The Cayley table of A_4 rendered with fiber figures. Each cell displays the figure associated with the product in that cell.

Internal verification identities

An implementation can verify its data using

$$\sigma_h \sigma_e^{-1} = \lambda_h$$

for all $h \in G$ and

$$|E(\Gamma_h)| = 2(n-1) - a(\sigma_h).$$

These identities are inexpensive and detect indexing or ordering errors early.

11 Further structural questions

The product-fiber formalism suggests several directions.

Classification of fibergraph multisets. For a fixed order, one may study the multiset of graph isomorphism types

$$\{[\Gamma_h] : h \in G\}.$$

Because the adjacency depends on the order, one can also minimize or average graph invariants over all orderings, producing order-independent optimization invariants of the group.

Ordering optimization. The quantity

$$\sum_{h \in G} a(\sigma_h)$$

measures the total overlap of the two Hamilton paths across all fibers. Maximizing it yields sparse fibergraphs; minimizing it produces more cyclic figures. Finding extremal orderings is a finite combinatorial optimization problem tied to the regular representation.

Conjugacy visualization. The diagonal $G \times G$ action sends F_h to $F_{xhx^{-1}}$. It therefore gives a direct geometric mechanism for visualizing conjugacy classes and central elements: the identity and central fibers are fixed at the label level under diagonal conjugation, while noncentral fibers move within their conjugacy classes.

Graph-directed limits with nonuniform ratios. If the contraction ratio depends on the outer state and child state, then the dimension is determined by the spectral radius of the weighted matrix

$$M(s) = (r_{h,k}^s)_{h,k \in G}$$

under the usual graph-directed separation hypotheses. This allows group-dependent dimensions once the geometry assigns genuinely nonuniform ratios.

12 Conclusion

Product fibers provide a compact bridge between Cayley tables, permutation representations, sparse graphs, and recursive geometry. The central algebraic identity

$$\sigma_h = \lambda_h \tau$$

shows that the fiber permutations form a coset of the regular representation, and the relative identity

$$\sigma_h \sigma_e^{-1} = \lambda_h$$

shows that the regular action can be recovered directly from the family of fibers.

The ordering-dependent edge rule also has a complete combinatorial description:

$$\Gamma_h = P_n \cup \sigma_h^{-1}(P_n).$$

Thus every fibergraph is the union of two Hamilton paths, with exact edge and cycle-rank formulas. This separates the intrinsic algebra from the chosen visual order and clarifies which aspects of the construction are structural and which are representational.

The action of G on fibers is canonical at the point-set level. The natural enlargement to $G \times G$ simultaneously contains left multiplication, right multiplication, and conjugation. Continuous animations should therefore transport vertices according to these canonical actions while treating the ordering-dependent edge layer separately.

Finally, the fiber family generates coupled graph-directed recursive systems. Uniform substitution has an all-ones state matrix and therefore a similarity dimension depending only on group order and scale. The permutation-labelled circle substitution is more algebraically informative: the child placed at the j th vertex of F_h is exactly $F_{hg_j^{-1}}$. In this way the multiplication law is encoded not only in the original Cayley table, but recursively in the spatial organization of the entire family.

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A A compact formula sheet

For convenience, the principal identities are collected here. With index map $\nu(g_i) = i$,

$$\begin{aligned}
F_h &= \{(i, j) : g_i g_j = h\}, \\
\sigma_h(j) &= \nu(h g_j^{-1}), \\
\lambda_h(i) &= \nu(h g_i), \\
\tau(i) &= \nu(g_i^{-1}), \\
\sigma_h &= \lambda_h \tau, \\
\sigma_h \sigma_k^{-1} &= \lambda_{hk^{-1}}, \\
\lambda_h &= \sigma_h \sigma_e^{-1}, \\
\Gamma_h &= P_n \cup \sigma_h^{-1}(P_n), \\
a(\sigma_h) &= \#\{j < n : |\sigma_h(j+1) - \sigma_h(j)| = 1\}, \\
|E(\Gamma_h)| &= 2(n-1) - a(\sigma_h), \\
\beta_1(\Gamma_h) &= n-1 - a(\sigma_h), \\
T_x(i, j) &= (\lambda_x(i), j), \\
T_{x,y}(i, j) &= (\lambda_x(i), \rho_y(j)), \\
T_{x,y}(F_h) &= F_{xhy^{-1}}, \\
\rho_y(j) &= \nu(g_j y^{-1}), \\
g_{\sigma_h(j)} &= h g_j^{-1}.
\end{aligned}$$

For a complete graph-directed state system with common ratio r and the open set condition,

$$\dim_H K_h = \frac{\log n}{-\log r}.$$